



Edmond Halley's isogenic chart shows lines of magnetic variation from true magnetic north.

AS THE 18TH CENTURY

DAWNED, British astronomer **Edmond Halley** sailed the Atlantic on his third **voyage of discovery**. In January 1700, he made the first observation of the **Antarctic convergence**, where icy Antarctic waters come up against warmer Atlantic waters in a ring around Antarctica. On February 1, he made the first recorded sighting of **tabular icebergs**, which have steep sides and a flat top. Halley also showed that **Earth's magnetism fluctuates** too much for compasses to be used to find longitude at sea. He confirmed that **magnetic north** does not correspond with **true north**, a phenomenon known as **magnetic declination** (see 1598–1604).

Also in 1700, French physician **Nicolas Andry** (1658–1742) suggested that **smallpox** was

JETHRO TULL (1674–1741)

Born in Berkshire, England, Jethro Tull intended to enter politics in London, but ill health kept him at home, farming. Noticing that hand-sown seeds were scattered chaotically, he developed the mechanical drill to sow seeds in even rows. He became a key figure in the agricultural reforms that swept through England in the 1700s and then around the world.



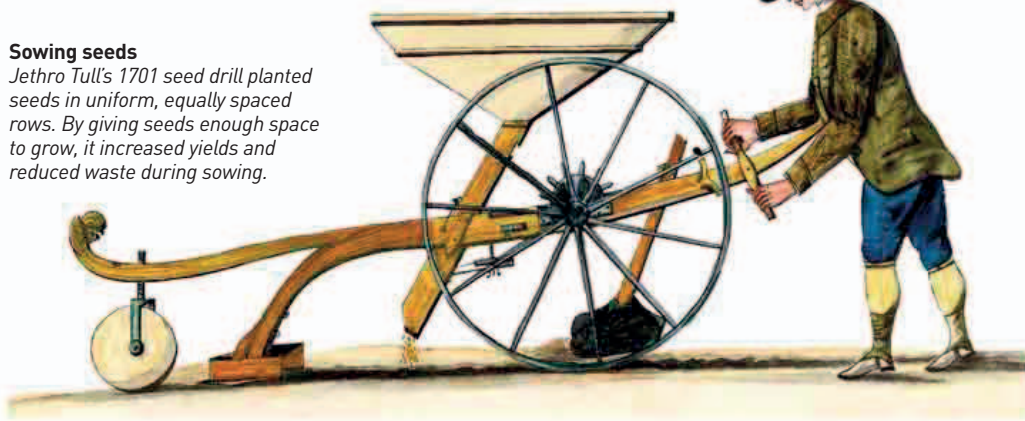
caused by **tiny microorganisms** or "worms" that he had seen through a microscope.

In 1701, English agriculturalist **Jethro Tull** helped modernize farming practices when he invented the **mechanical seed drill**—a machine that

automatically planted seeds in neat, evenly spaced rows. The adoption of Tull's method increased crop yields by as much as 900 percent.

Sowing seeds

Jethro Tull's 1701 seed drill planted seeds in uniform, equally spaced rows. By giving seeds enough space to grow, it increased yields and reduced waste during sowing.



Isaac Newton, here seen speaking at a meeting of the Royal Society, was elected its president in 1703.

ISAAC NEWTON'S SCIENTIFIC ACHIEVEMENTS

became increasingly well known in the 18th century. A key moment in the growth of his fame was the 1702 publication of *Astronomiae Physicae et Geometricae Elementa* (*Elements of Astronomy, Physics, and Geometry*) by Scottish mathematician David Gregory. One of the first popular accounts of Newton's theories, this work discussed his ideas on gravity and the movement of the planets. Newton was **elected president of the Royal Society**

By the 18th century, scholars were starting to consider natural events, such as earthquakes, as phenomena to be investigated scientifically rather than as acts of God. In 1703, French priest and inventor **Abbé Jean de Hautefeuille** (1647–1724) described a **seismometer** for measuring the severity of earthquakes. De Hautefeuille's device, a simple balanced pendulum whose swing responded to ground movement, was one of the earliest seismometers used in Europe.

500,000

THE NUMBER OF EARTHQUAKES THAT OCCUR EACH YEAR

in London in 1703, a post he held until his death in 1727.

In 1703, German chemist **Georg Stahl** developed Johann Becher's 1667 idea that an element called *terra pinguis* is released from substances such as wood when they burn. Stahl called the element phlogiston, and the **phlogiston theory of combustion** came to dominate 18th-century chemistry until finally disproved by Antoine Lavoisier (see 1789) later in the century.

Botanists, meanwhile, were beginning to embark on voyages of exploration to study the rich variety of unknown plants in newly discovered parts of the world. After three plant-hunting voyages to the West Indies, French botanist **Charles Plumier** published *Nova Plantarum Americanarum Genera*, a huge and groundbreaking work on **plant classification**. In it, he described the plants fuchsia and magnolia for the first time.

1700 French physician **Nicolas Andry** suggests smallpox is caused by tiny "worms"

January 1700 English astronomer **Edmond Halley** observes the Atlantic convergence

1700 Italian physician **Bernardino Ramazzini** publishes *Diseases of the Workers on occupational medicine*

1701 Jethro Tull develops the seed drill

1703 German chemist **Georg Stahl** devises phlogiston theory

1703 French priest Abbé **Jean de Hautefeuille** invents a seismometer

1703 French botanist **Charles Plumier** describes the plants fuchsia and magnolia